

Project Proposal

AI-Powered Smart Tourism Assistant Platform with Personalized Trip Planning

Group No 08

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Overview of the Project

This project focuses on resolving the highly fragmented and time consuming methods of modern travel planning. Currently travelers must navigate multiple platforms for itineraries, weather, and local guidance which relies on general AI tools that lack curated and trusted data.

This project introduces a trusted and centralized **AI-Powered Smart Tourism Assistant Platform** which is driven by Agentic AI and anchored by a trusted, curated database. This platform eliminates information fragmentation to deliver highly reliable, personalized trip-planning experiences within a single platform.

The project will deliver a fully integrated platform consisting of the following core components:

- **User-Facing Mobile & Web Applications:** Cross-platform applications featuring conversation-based user interface designed for seamless traveler interaction.
- **Agentic AI System:** A specialized backend AI engine built to autonomously generate, refine, and optimize travel plans based on user preferences.
- **Admin Panel:** A dedicated management interface used to input, verify, and maintain the trusted internal database of hotels, restaurants, local events, and popular attractions.

System Inputs

- User registration and profile information
- User queries submitted through the chat interface
- Curated Internal Data (Verified, reliable details on travel destinations, and events)
- Real-Time Contextual Data (Ex: live weather updates and forecasts)
- User Preferences (Travel style, budget constraints etc.)

System Outputs

- Personalized travel itinerary
- Suggested hotels, restaurants and attractions and weather information
- Budget estimation
- Usage reports and analytics
- Interactive travel destinations map (mobile)

The platform is also planned to be deployed on a cloud environment using AWS, allowing users to access the system reliably while supporting future scalability and maintenance.

Objectives of the Project

The objectives of this project are to:

- Deliver a functional, complete, centralized smart tourism platform (mobile and web application).
- Analyze travel planning challenges.
- Implement personalized itinerary generation.
- Develop an Agentic AI System for goal-oriented trip management using curated data.
- Provide budget planning and optimization.

The Need for the Project

- **Eliminating Information Fragmentation:** Instead of searching through multiple websites and apps for schedules, weather, and guides, this platform centralized the entire planning process into a single conversational interface.
- **Providing curated and trusted data:** Uses an admin-verified database to guarantee that all recommendations for hotels, restaurants, and events are completely reliable.
- **Intelligent Itinerary Optimization:** Balancing a limited timeline, a strict budget, and personal interests is tough. The platform automatically handles this logistics puzzle, creating customs, optimized schedules that save both time and money.

Scope of the Project

The proposed system consists of four main components, delivered across both a web application and a mobile application (Android and iOS).

1. Tourist / User

Available on both web and mobile, the user will be able to:

- Register and log into the system
- Create and manage travel plans
- Interact with the AI assistant using a conversational, chat-like interface
- Receive personalized itinerary recommendations
- Search and filter attractions, restaurants, and events
- View nearby attractions using an interactive map (mobile)
- Receive travel notifications (mobile)
- View current weather information
- Access previous trips and travel history

2. Administrator

Through the Admin Panel, the administrator will be able to:

- Manage tourist attractions, restaurants, hotels, and local events
- Update destination information
- Monitor user activity and system usage
- View system analytics and reports (web)
- Manage all content displayed to users

3. Agentic AI System

Operating exclusively on data verified through the Admin Panel, the Agentic AI System will:

- Understand user travel requirements and goals from natural-language input
- Generate and optimize personalized itineraries
- Recommend attractions according to user interests
- Optimize travel schedules across the trip duration
- Estimate and optimize travel budgets
- Retrieve and present real-time contextual data, including weather and local events

4. System Features

- User authentication (JWT-based)
- REST API services
- AI-powered chat interface
- Interactive maps (Mobile)
- Weather integration
- Notification services
- Data synchronization across web and mobile
- Reports and analytics
- Cloud deployment using AWS for reliable and scalable system hosting.

Deliverables

The project will deliver:

- A responsive, web-based tourism platform
- A mobile application for Android and iOS
- A functional Agentic AI travel-planning assistant
- An administrator dashboard
- RESTful API services
- A database management system

- Project documentation
- API documentation
- Final project report

Overview of Existing Systems and Technology

While several platforms address specific travel needs, none provide a complete, verified solution for personalized planning:

- **Google Travel:** Offers strong mapping and itinerary management but relies on broad global data rather than locally curated content.
- **TripAdvisor:** Specializes in crowd-sourced reviews but lacks automated, personalized AI itinerary generation.
- **Booking.com:** Focuses primarily on reservations, leaving itinerary planning highly limited.
- **General AI (ChatGPT, Gemini, Claude):** Provides conversational advice but lacks access to a verified, destination-specific database, limiting the reliability of its suggestions.

The Proposed System bridges these gaps by combining a conversational AI interface with an admin-managed, trusted local database. This ensures itineraries are not only hyper-personalized but also highly reliable, eliminating the fragmentation and uncertainty of current tools.

Technologies Planned for Implementation

Component	Technology
Backend	NestJS
Frontend (Web)	Next.js
Mobile Application	Flutter
Database	PostgreSQL
Authentication	JWT Authentication
AI Service	Python, FastAPI, LangChain, Gemini API
Maps	Google Maps API
Weather	OpenWeather API
Notifications	Firebase Cloud Messaging
API Communication	REST APIs
Version Control	Git & GitHub
Cloud Deployment	AWS (Amazon Web Services)

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